Major Project Phase I Report

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Mentor: Vivek Menon Sir
Made by:-
Satyesh Uikey
Srijan Singh
Sudhanshu Chouksey

About The Project

This project aims to develop an AI-powered healthcare assistant that supports both voice and text communication, allowing users to interact in the language they are most comfortable with. The system provides basic medical guidance by analyzing symptoms, recommending possible lab tests, and helping users understand their health conditions more clearly. Built with the intention of making healthcare information more accessible, the chatbot uses cloud-based infrastructure to securely store user interactions and ensure smooth operation. By combining multilingual capabilities, natural language processing, and an intuitive interface, the project strives to bridge the gap between users and reliable preliminary medical support.

Overview

Project Name: Healthy-Partner

Repository:

Development Type: Full Stack Web Application with Artificial Intelligence

Target Users:-

- Rural households
- Healthcare workers
- NGOs and government agencies
- Policy Holders
- General Public

The healthcare chatbot uses the MedAlpaca medical NLP model to process symptoms and generate meaningful responses based on user input. It supports multimodal communication by accepting both text and voice messages, converting spoken input into text, and analyzing it through AI. The backend is developed using Python's FastAPI framework and deployed on AWS to ensure reliability, scalability, and data protection. All user conversations and uploaded files are securely stored using AWS services, enabling users to revisit past health queries when needed. The system focuses on providing safe, accurate, and easy-to-understand medical insights, serving as an accessible first level of healthcare assistance.

Our Services

The system provides a range of AI-powered healthcare services designed to guide users in understanding their symptoms and taking the right next steps. It enables natural communication through multilingual voice and text, offering users a personalized and accessible interaction experience. The chatbot also identifies symptoms from user input and provides preliminary medical guidance along with suggestions for essential laboratory tests. In addition to offering treatment-related insights, the system keeps users’ records safe and organized through secure storage, allowing them to maintain a clear history of their health-related conversations.

Tech Stack

Web Framework and Server

- Python(FastAPI)
- Uvicorn

Document Processing and OCR

- unstructured[pdf, ocr]
- pypdf

Frontend (for local testing)

- streamlit

Flask Backend & Twilio Integration

- Flask
- Twilio

OCR for Medical Images

- easyocr

Frontend Implementation

The frontend follows a component-based structure and includes:

- Home Page
- User Registration
- Login Page
- Health Analysis Page
- Policy Analysis
- Report Analysis
- AI Chat Assistant
- Contact Page

Features implemented:

- Dark mode
- Responsive UI
- Mobile optimization
- Real-time form validation
- Toast notifications
- File uploads

Backend Implementation

The backend implements:

- MVC architecture
- Secure user authentication
- RESTful APIs
- AI service integration
- Image processing
- Token-based authorization
- Role-based scalability

All APIs follow:

- Structured input validation
- Controlled error handling
- Secure cookie management

The Database

Users Collection
Stores:

- Personal info
- Policy data
- Authentication details
- Timestamps

Tokens Collection
Stores:

- Refresh tokens
- Expiry time
- User references

Indexes:

- Email
- User ID
- Expiry

AI Integration

Google Gemini 2.0 Flash

Vision-based image analysis

NLP-based responses

Prompt engineering for:

- Healthcare
- Policy Management
- Report analysis

Conclusion

This project demonstrates how multilingual AI, speech processing, and cloud technologies can be combined to create an accessible and effective healthcare assistant. The system supports users in understanding their symptoms, exploring possible health concerns, and accessing preliminary medical guidance without barriers of language or technology. By providing a scalable, secure, and user-centric solution, the chatbot contributes to a more inclusive digital healthcare environment. It sets the foundation for future advancements such as telemedicine integration, enhanced diagnostics, and predictive health analytics.